

Modelling with linear and quadratic functions

Turning a paragraph of Uzbek life into a function, solving it, and then checking the answer means something.

LESSON 24–25

2 × 40 MIN

ALGEBRA 10, §1.8

P1 · 1.7, 3.5

LESSON CLOCK

8'

16'

22'

16'

20'

8'

The four-step cycle	8 min
Linear models	16 min
Quadratic models and the vertex	22 min
Judging a model	16 min
Practice	20 min
Homework	8 min

LEARNING OBJECTIVES

- Define variables clearly and write a function from a described situation.
- State the domain that the context — not the algebra — allows.
- Solve the model and interpret the solution in the original words.
- Judge whether a model is reasonable and name one of its limitations.

TEXTBOOK

National	pp. 81–88
Cambridge	Pure Mathematics 1, pp. 18–23, 75–79
Workbook	P1 Exercise 1F, 3D

01

Explanation

The four-step cycle

NEVER SKIP STEP 1

1 Define — “let x be the width in metres”. **2 Translate** — build the function and state the domain the situation allows. **3 Solve** — ordinary algebra. **4 Interpret** — answer in the words of the question, with units.

Most lost marks in modelling questions are lost at step 1 and step 4, not step 3. A number with no units is not an answer, and “ $x = 4$ ” does not tell a reader what is 4 metres long.

Linear models — a fixed part and a rate

Any situation with a starting amount and a constant rate of change is linear:

$$y = (\text{starting amount}) + (\text{rate}) \times (\text{number of units})$$

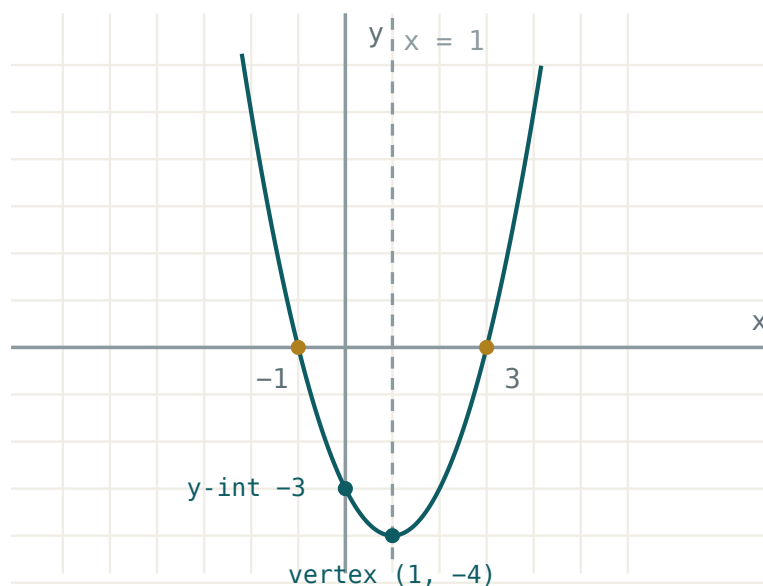
SITUATION	MODEL	WHAT C AND M MEAN
taxi: 8 000 so'm plus 2 500 a km	$C = 8000 + 2500d$	flag fall; price per km
a 40 000 so'm card losing 300 a minute	$B = 40000 - 300t$	credit; cost per minute
a tank filling at 12 litres a minute from 50 litres	$V = 50 + 12t$	start volume; flow rate

THE DOMAIN COMES FROM THE WORLD

$B = 40000 - 300t$ is algebraically fine at $t = 500$, but the card empties at $t = 133\frac{1}{3}$. The domain is $0 \leq t \leq 133.3$, and saying so is worth a mark.

Quadratic models — where the vertex is the answer

A quadratic appears whenever two quantities that vary must be multiplied — an area from two lengths, a revenue from price times quantity, a height under gravity.



For a negative parabola the vertex is the maximum. Its x tells you what to choose; its y tells you what you get.

Fencing. 60 m of fence, a rectangular pen against a wall (three sides fenced). Let the two equal sides be x . Then the third is $60 - 2x$ and

$$A(x) = x(60 - 2x) = 60x - 2x^2 = -2(x - 15)^2 + 450$$

so the greatest area is 450 m^2 when $x = 15$ — and the domain is $0 < x < 30$, because outside it a side would be negative.

WHICH COORDINATE ANSWERS THE QUESTION?

“What width gives the largest area?” wants the x of the vertex. “What is the largest area?” wants the y . Read the question twice before writing the number.

Judging the model

Every model makes assumptions. Naming one is part of a complete answer:

MODEL	AN ASSUMPTION	WHERE IT FAILS
taxi fare	the rate never changes	night tariff, waiting time
ball thrown up	no air resistance	light or fast objects
revenue = price × sales	sales fall linearly with price	very high or very low prices

A model is not right or wrong. It is useful over some range of inputs, and the honest answer says which range.

02 Worked examples

EXAMPLE 1 A rectangle has perimeter 40 cm. Find the greatest possible area.

① Let the width be x cm.

Length $20 - x$.

② $A(x) = x(20 - x) = 20x - x^2$

Domain $0 < x < 20$.

③ $= -(x - 10)^2 + 100$

Vertex at $x = 10$.

④ A square, side 10 cm.

ANSWER

100 cm², when the rectangle is a 10 cm square

EXAMPLE 2 A shop sells 200 notebooks a week at 12 000 so'm. Each 1 000 so'm rise loses 10 sales. What price maximises revenue?

① Let there be x rises of 1 000.

Price $12000 + 1000x$, sales $200 - 10x$.

② $R(x) = (12000 + 1000x)(200 - 10x)$

③ $= 2400000 + 80000x - 10000x^2$

④ $= -10000(x - 4)^2 + 2560000$

$x = 4$.

ANSWER

Price 16 000 so'm; revenue 2 560 000 so'm

EXAMPLE
3**A ball is thrown up: $h(t) = 20t - 5t^2$ metres after t seconds. Find the greatest height and when it lands.**

① $h(t) = -5(t - 2)^2 + 20$
Complete the square.

② Greatest height 20 m at $t = 2\text{ s}$.

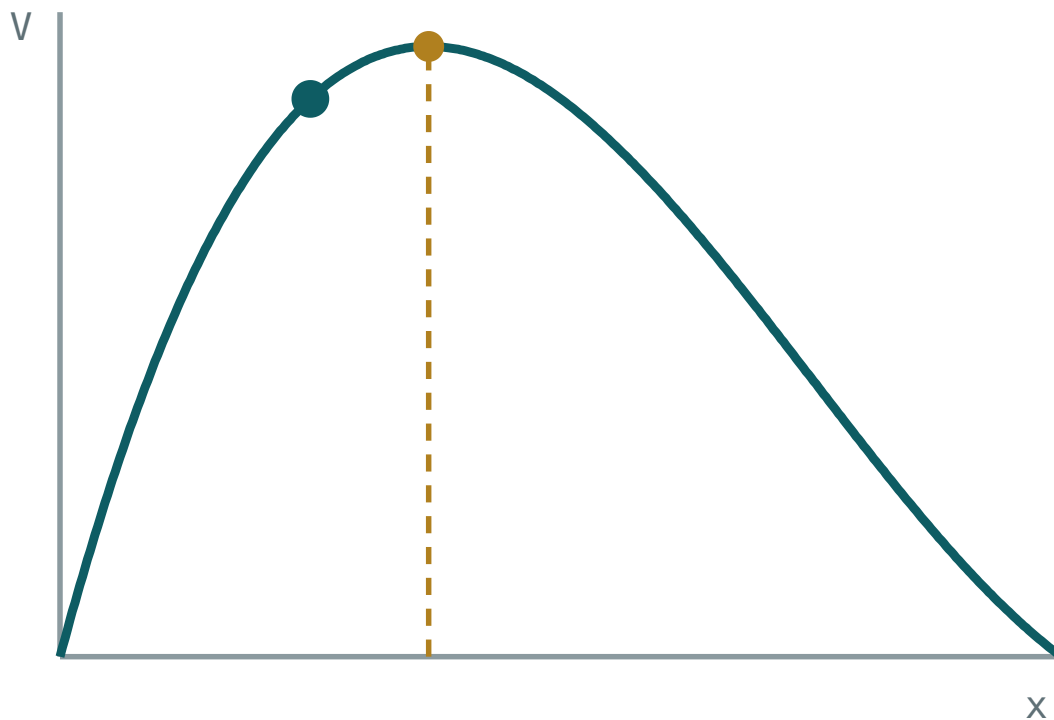
③ $20t - 5t^2 = 0 \Rightarrow 5t(4 - t) = 0$

④ $t = 0$ or $t = 4$.
Domain $0 \leq t \leq 4$.

ANSWER 20 m at $t = 2$; lands at $t = 4\text{ s}$ **03 Interactive model**

Change the total length of fence and watch the optimum width move with it.

The largest box

 $x = 2$ $V(x) = 384$ greatest $V = 420.1$ at $x = 2.94$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Mathematical model	Matematik model	Математическая модель
Variable	O'zgaruvchi	Переменная
Constraint	Cheklov	Ограничение
Constant term	O'zgarmas had	Свободный член
Rate of change	O'zgarish tezligi	Скорость изменения
Maximum value	Eng katta qiymat	Наибольшее значение
Optimal value	Optimal qiymat	Оптимальное значение
Interpretation	Talqin	Интерпретация
Assumption	Faraz	Допущение
Limitation of a model	Model cheklovi	Ограниченность модели

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In $C = 8000 + 2500d$, the number 2500 is:

the fixed charge

the rate per km

the total

the domain

For a negative quadratic, the vertex gives:

the minimum

the maximum

a root

the intercept

A rectangle of fixed perimeter has greatest area when it is:

long and thin

a square

a golden rectangle

it has no maximum

The domain of a length variable is normally:

$$\mathbb{R}$$

$$x > 0$$

$$x \neq 0$$

$$x \leq 0$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A taxi charges 9 000 plus 3 000 a km. Write $C(d)$

ANSWER $C = 9000 + 3000d$

2. Cost of 7 km on that model

ANSWER 30 000 so'm

3. A rectangle is x by $10 - x$. Write the area

ANSWER $A = 10x - x^2$

4. Domain of that area function

ANSWER $0 < x < 10$

5. $h(t) = 10t - 5t^2$; when is $h = 0$?

ANSWER $t = 0, 2$

6. Greatest value of $A = -(x - 6)^2 + 36$

ANSWER 36

7. Where does it occur?

ANSWER $x = 6$

Medium · the standard question

7 PROBLEMS

1. Perimeter 24 cm; maximise the area

ANSWER 36 cm^2 , a 6 cm square

2. A card holds 25 000 and loses 400 a minute. When is it empty?

ANSWER 62.5 minutes

3. $h(t) = 30t - 5t^2$; greatest height

ANSWER 45 m at $t = 3$

4. Two numbers add to 20. Maximise their product

ANSWER 100, at 10 and 10

5. Fence 100 m, pen against a wall. Maximise the area

ANSWER 1250 m^2 with $x = 25$

6. A firm sells $100 - 2p$ items at price p . Write the revenue

ANSWER $R = 100p - 2p^2$

7. Maximise it

ANSWER $p = 25$, $R = 1250$

Hard · stretch and reason

7 PROBLEMS

1. Fence 80 m against a wall, but a 4 m gate is free. Maximise the area

ANSWER $x = 21$, area 882 m^2

2. Cost $C = x^2 - 40x + 500$. Find the least cost and the x

ANSWER 100 at $x = 20$

3. $h(t) = 1.5 + 12t - 5t^2$. When does the ball reach 8 m?

ANSWER $t \approx 0.71$ and $t \approx 1.69 \text{ s}$

4. Two numbers differ by 6. Minimise the sum of their squares

ANSWER 18, at 3 and -3

5. A 20 cm wire is cut in two and each part bent into a square. Minimise the total area

ANSWER cut in half; 12.5 cm^2

6. Sales fall by 5 for every 500 so'm rise from 10 000 where 300 sell. Maximise revenue

ANSWER price 20 000, revenue 3 000 000

7. Name one assumption in the previous model and where it fails

ANSWER Sales fall linearly — false at very high prices, where they hit zero

07 Homework

Homework — 5 tasks

Every answer needs units and a sentence, not just a number.

1. A plumber charges 50 000 so'm plus 30 000 an hour. Write the model, find the cost of a $3\frac{1}{2}$ hour job, and state the domain.
2. A rectangular garden has perimeter 36 m. Find the dimensions giving the greatest area.
3. $h(t) = 25t - 5t^2$. Find the greatest height, the time it occurs and when the ball lands.
4. A stall sells 80 samsa a day at 8 000 so'm. Each 500 so'm rise loses 4 sales. Find the revenue-maximising price.

5. Name one assumption in task 4 and say where it would break down.